* Algorithm → Search | Sort

↳ search → Coding way to find specific value
- → linear search
- → Binary search

* linear search →

* → check the search value with each element in four Array

→ Adv → very simple
→ Can work on un ordered Array

→ disadvantage
↳ very slow with large Array

Note
① → own → Based on size
② Big of o → Big o
↳ describe → execution / space

* ex:. write c. code to find Index (Array) for Input value.

```
#include <stdio.h>
int main()
{
  char No [5]={0};
  char count =0;
for (count=0; count<5; count++)
  {  scanf("%hhd", &No[count]);
  }
  printf("Enter search value:");
  char searchvalue =0;
  scan ("%hhd", &search value);
```

// linear

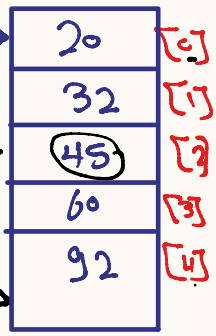
```
for (count=0; count<5; count++)
{  if (No [count] == search value)
   {  printf("%d", count);
      break; → // key word to terminate the loop
   }
}
```

* Big of o
↳ Mathamtical equation
↳ to describe the
↳ execution time
↳ space complexity
↳ like →
↳ O(1) → Constant time
↳ O (nlogn) → linearithmic time
↳ O(n²) → Quadratic time

* Return 5:10
↳ write C_Code to get the No of Repeate for Input No from user in Input Array

* searching → Binary
↳ use to find specific element in ordered array
↳ O(logn) → Based on Array divid

ex!. array [5] →



$Mid = \frac{start + End}{2} = \frac{0 + size-1}{2}$

$Mid \Rightarrow \frac{0+4}{2} = 2$

search value == array [Mid]

search value > array [Mid]

search value < array [Mid]

find value

find value example (44)

* start=0; End=5;

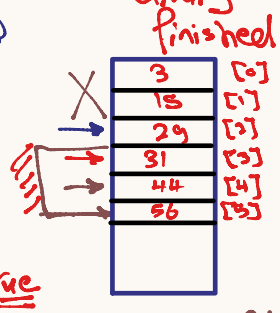
$Mid = \frac{0+5}{2} = 2$ (29)

value == arr [Mid] → x

value > arr [Mid] → true

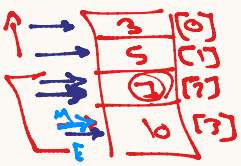
* start = Mid+1 → 2+1

End=5; New Mid = $\frac{3+5}{2}$ Mid=4



ex2:. Search of value Not exist

int arr [4] →
search value = 6



① start = 0
② End = 3
③ Mid = 1

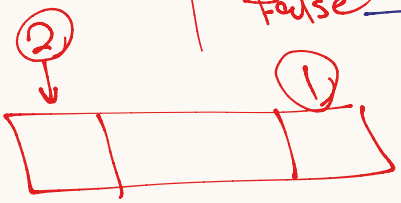
value == arr [Mid]
6 5
false

value > arr [Mid]
6 5
start = Mid + 1
End = Fixed.
start = 2
End = 3
Mid = 2+3/2 = 2

value == arr [mid]
false

value > arr [mid]
6 7
false

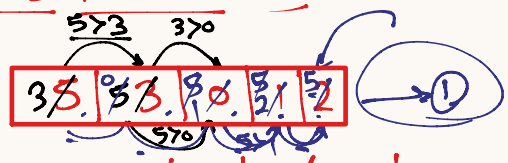
value < arr [mid]
End change
start
start = 2
End = Mid - 1
= 2 - 1
End = 1



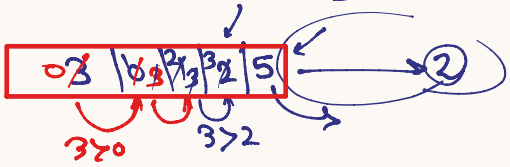
* still search on value until the find element or End > start

* write C_ Code to ask the user to Enter the ordered temp value
* and ask him to Enter the best temp degree → and print the Index for this temp

* Sorting ×1 Bubble Sort



① Ask element with element for all element to get Max No and set it in the last



① How to swap

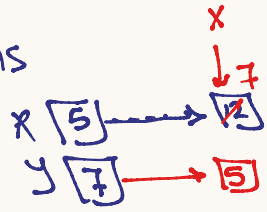
[1] by using temp value

```
int temp = x;
x = y;
y = temp;
```

x [5] [7]
y [7] [5]
t [5]

[2] by Add & Mins

```
x = x + y;   // 12  5  7
y = x - y;   // 5   12 7
x = x - y;
```



[3] by XOR (^)

```
x = x^y;
y = x^y;
x = x^y;
```

* → Two nested for loop

```
for (i = 0; i < size-1; i++)        // 1
    for (j = 0; j < size-1-i; j++)  // 2
        if (Array[j] > Array[j+1])
            Swap
```

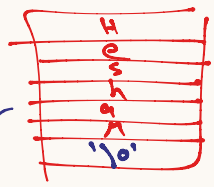
* string

↳ Array of char

* char Name [ ] = "Hesham";

string

Compiler add: '\0' : Null char



* Size of string = Number of char + 1

* Print string

```
char Name[ ] = "Hesham";
Printf("%s", Name);
```

string

* Scan →

```
char Name [20] = {0};
Scanf("%s", Name);
```

Array Name → Address

Modifier

* → signed → +ve, -ve

Compiler

Variable: store +ve only → unsigned

unsigned char No;
unsigned int No;

* int * char only

Variable: store +ve & -ve → signed

signed char ↕
signed int ↕

if you don't select Compiler will be select int x

char → 1 Byte

unsigned Char

0000 0000 → 0
1111 1111 → 255

Range No
0 : 255 → +ve

signed Char

↳ 0000 0000

↳ 1111 1111

Rang
0 : 127 +ve
-1 : -128 -ve

ex:

short int x = 30; → 2 Byte
long int y = 40; 4B → 4 Byte
long long int z = 60; → 8B
8 Byte

* C version → C89, C90, C99, C11, C

short int → 2 Byte
long → 4 Byte
int → 4 Byte
long int → 8 Byte
long long int → 8 Byte

* Size Modifier

char → at least 1 Byte
int → at least 2 Byte
↳ 2, 4, 8 Byte
Compiler → System Bus

float → at least 4 Byte
double → at least 8 Byte

Modifier

① short → int
Compiler → 2 Byte
short int

② long → int 4 Byte
↳ double

③ long long → int 8 Byte

unsigned short int timer = 0;

Data type

* typedef → Rename for Data type → Compiler

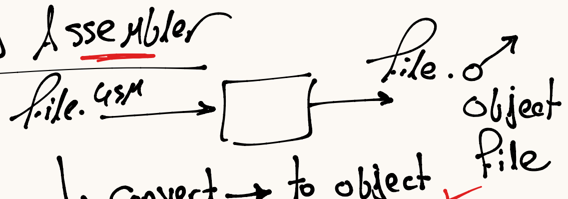
typedef old Name New Name ;

ex:. typedef unsigned short int Hamada ;

* create Header file
* stdtype.h

typedef unsigned char U8; (unsigned, No of Bit)
typedef signed char S8; (signed, No. of bit)

* [3] Assembler



↳ Convert → to object file
↳ Provid. symbol table

	x	Provid
	printf	need

global var
function
x →

file1.c → file1.o

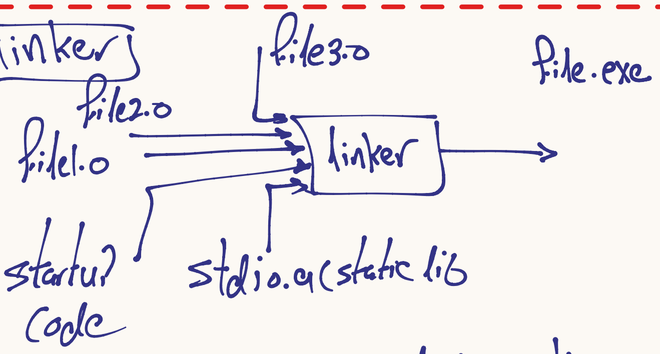
```
#include <stdio.h>
int x = 20;
int main() ←
{
printf("%d \n", Add(5,7));

}
```

symbol table

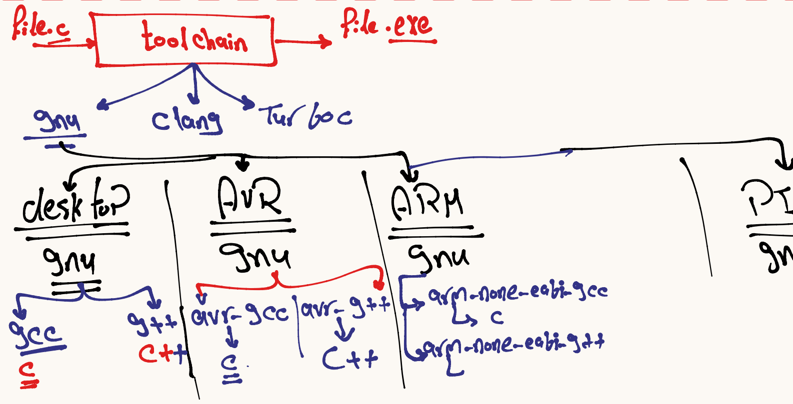
s/	st
x	Pro
Main	Prov
Pr	Need
Add	Need

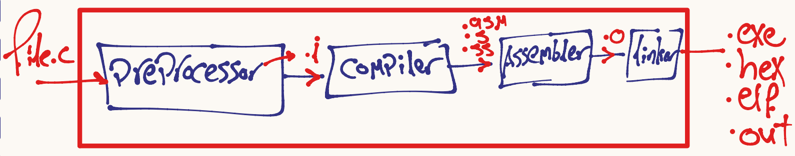
[4] linker



* → linker will be link with different files by symbole table

* → Collect all object file in one file executable
↳ give Physical Address





[1] PreProcessor

file.c → [1] → file.i

* text Replacement for PreProcessor directives.

#include #define #undef
#ifndef #ifdef #if #else
#error #warning #elif #endif

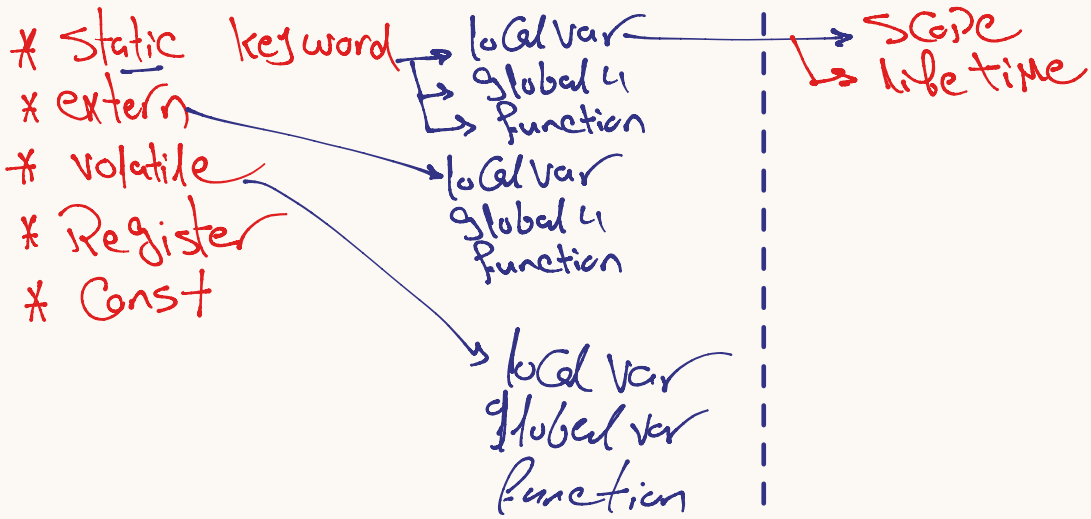
gcc -E file.c -o file.i
↳ flag → PreProcessor only

[2] Compiler

file.i → [ ] → file.asm
↳ check syntax error
↳ check some logic error
↳ optimize
↳ Convert to Assembly

gcc -S file.c -o file.asm

* Notes
(1) #pragma
↳ Compiler directives
#pragma once


* Static keyword
* extern
* volatile
* Register
* Const
local var
global &
function
local var
global &
function
local var
global var
function
scope
lifetime